# LLM Limitations and Safety in Agent Collaboration

A case study in **reliable** and **verifiable** hypothesis generation

Mingxue Xu

mingxue-xu.github.io

7 October 2026, Oxford Collaboration Week

# Outline


- **0. Scientific creativity and hypothesis generation**
- **1. Limits of individual models**
  - Resource and capacity limits
  - Managing context in long tasks
  - Nondeterminism in deployed LLM systems
- **2. Generating and checking hypotheses with agents**
  - Agent roles in GraphHypoth
  - GraphHypoth architecture overview
  - Checking claims against the literature
  - How GraphHypoth calculates scores
  - Claim graphs and unsupported claims

# Scientific creativity and hypothesis generation

# Scientific creativity and hypothesis generation

**Scientific reasoning**

Scientific reasoning involves forming hypotheses (problem finding), testing them (problem solving), and drawing logical inferences.

- **Creative ideas** are both *new* and *useful*.
- **Where ideas come from:**
  - **Within a field:** deep knowledge helps researchers notice new connections.
  - **Across fields:** ideas from one field can suggest new approaches in another.

Abraham (2018)

© 2026 Mingxue Xu

# 1. Limits of individual models

# 1.1 Resource and capacity limits

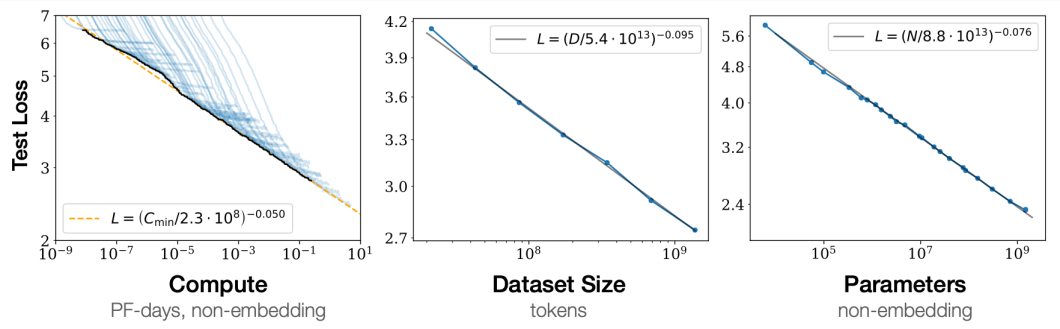
Research on transformer language models examines several kinds of limits:

1. **Scaling laws.** Kaplan et al. (2020) measure how loss changes with parameters, data, and compute. Hoffmann et al. (2022), in Chinchilla, study how to divide training compute between model size and data: $\hat{L}(N, D) = E + A/N^{\alpha} + B/D^{\beta}$ under $C \approx 6ND$.
2. **Expressivity.** Which functions can a network represent (Xu et al., 2024)? Theoretical limits depend on the architecture. Hahn (2020) shows that fixed-depth hard-attention transformers cannot recognise parity over arbitrary lengths.
3. **Capacity.** How much information can a network memorise (Xu et al., 2024)? Allen-Zhu and Li (2024) relate recoverable knowledge to bits stored in weights. Their estimate of $\approx 2$ bits/parameter is empirical.

Allen-Zhu and Li (2024); Hahn (2020); Hoffmann et al. (2022); Kaplan et al. (2020); Xu et al. (2024)

© 2026 Mingxue Xu

# Figure A. Loss as compute, data, and model size increase




In these experiments, test loss falls approximately as a power law as compute, data, or parameter count increases.

Kaplan et al. (2020), Figure 1

© 2026 Mingxue Xu

# 1.1 What these limits mean for agent design

- A model's limits depend on its task and architecture, numerical precision, training resources, and inference procedure.
- Storage bound: $P$ parameters at $b$ bits each allow at most $2^{Pb}$ configurations. Recovering $B$ independent binary facts exactly requires $B \leq Pb$. This bound does not cover compressible facts or reasoning accuracy.

**Task design and checks**

Give each agent a *clearly defined task* and evaluate its outputs. Security controls must still work when the agent makes a mistake.

© 2026 Mingxue Xu

# 1.2 Managing context in long tasks

**What engineering reports show**

Long histories accumulate irrelevant messages and tool outputs. Agents need context relevant to the next step.

Reset strategies depend on the model. Anthropic removed manual resets when moving from Sonnet 4.5 to Opus 4.5.

Retest context management after model upgrades.

**Practices for long tasks**

- **Load sources when needed.** Keep references and retrieve the relevant passages.
- **Save progress.** Record decisions and unfinished work outside the conversation.
- **Shorten the history carefully.** Remove repetition but keep details the task still needs.
- **Keep tasks focused.** Give each specialist a defined task and ask for a brief report.

**Proposed use in GraphHypoth**

Pass the claim, evidence IDs, verification status, and next action to the next agent. Keep permission checks in code outside the conversation.

Rajasekaran et al. (2025); Martin et al. (2026)

© 2026 Mingxue Xu

# 1.3 Nondeterminism in deployed LLM systems

**Why runs differ**

**Numerical effects.** Even at temperature 0, outputs can differ when batching changes floating-point results.

Thinking Machines demonstrated repeatable inference with kernels that preserve results across batch sizes.

**Execution environment.** CPU and memory limits, along with timeouts, can change agent outcomes and benchmark scores.

**Evaluating repeated runs**

- **Record the setup.** Log the model, prompts, tools, data, and resource limits.
- **Repeat trials.** Reset the environment and distinguish infrastructure failures from task failures.
- **Review the runs.** Inspect failures and check that the graders work as intended.
- **Report consistency.** Include success per trial and the chance that all $k$ trials succeed ($\text{pass}^k$).

**Success across repeated trials**

If independent trials each succeed 75% of the time, all three succeed with probability $0.75^3 \approx 42\%$.

He and Thinking Machines Lab (2025); Grace et al. (2026); Segato (2026)

© 2026 Mingxue Xu

# 2. Generating and checking hypotheses with agents

# 2.1 Agent roles in GraphHypoth

GraphHypoth uses seven agent roles to generate hypotheses, check the evidence, and plan experiments. The final role writes up the hypotheses.

**Extraction Agent:** Turns the initial claim or research goal into concept nodes and directed edges. Each edge starts as unverified.

**Evidence Reviewer:** Grades the shortlisted passages for each edge. Code combines the grades into a status and confidence score, then saves both on the edge.

**Research Synthesist:** Works in one ongoing thread. It finds concepts in retrieved passages and proposes hypotheses using the saved graph, then revises them after review. Fixed rules filter and rank the candidates.

**Critic Panel:** Grades and ranks proposed hypotheses, then returns feedback to the Synthesist. Two judges always take part. A third joins when they disagree.

**Experiment Designer:** Drafts a plan for each confirmed hypothesis once it is saved to the graph. It uses retrieved methods and the graph's confounder and mediator nodes.

**Experiment Validator:** Grades the plan on five criteria, with one revision round by default. Once the plan includes all required content, the workflow saves it against the hypothesis edge.

**Narrative Elaborator:** Writes descriptions of the hypotheses for the run reports. It runs by default. The `-no-elaborate` flag turns it off.

Source: Mingxue-Xu (n.d.-a), GraphHypoth: README, Agent Roles and What A Complete Run Does. Retrieved October 13, 2026.

© 2026 Mingxue Xu

**Shared claim graph**

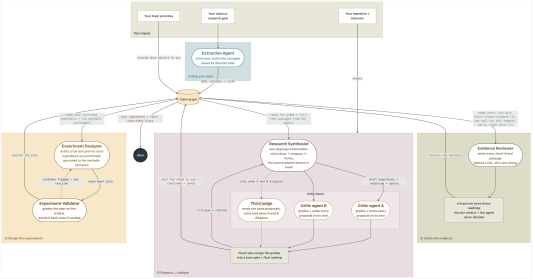
- Nodes: concepts
- Directed edges: relationships asserted by sources

**Stages**

1. Extraction
2. Evidence retrieval and review
3. Hypothesis development
4. Critique
5. Experiment planning

**Model judgments and code**

LLMs provide judgments. Code applies fixed scoring rules and controls state updates.






Facthink (n.d.-b); Mingxue-Xu (n.d.-a)

# 2.2.1 Checking claims against the literature

- **RAG** (Lewis et al., 2020) uses retrieved sources to help generate an answer.
- **FActScore** (Min et al., 2023) splits an answer into individual factual claims and checks whether a knowledge source supports each one. Its automated estimator uses an LLM.

**Checking research claims**

- Link each claim to the relevant passage in a source.
- Check whether the passage supports the claim, including its qualifications.
- Check the citation itself: the source may be irrelevant or misinterpreted.
- Treat retrieved text as *untrusted input* and handle it separately.

Lewis et al. (2020); Min et al. (2023)

© 2026 Mingxue Xu

# 2.2.2 How GraphHypoth calculates scores

**Evidence association**

$$A = 0.55(0.8E + 0.2L) + 0.35C + 0.10P$$

$E$: embedding similarity, $L$: lexical match, $C$: citation overlap, $P$: source prior.

**Edge confidence for verdict $v$**

$$C_v = \mathrm{clip}_{[0,1]}(0.55T_v + 0.25B_v + 0.20D_v - 0.25R - 0.15Q)$$

$T_v$: top-three evidence mean, $B_v$: support balance, $D_v$: source breadth, $R$: risks, $Q$: qualifiers (counts capped after division by 3).

**Hypothesis score and final rank**

$$H = \mathrm{clip}_{[0,1]}\left(\frac{N + P_h + T_h + Y + G + M}{6}\right)$$

$$\mathrm{RankScore} = \mathrm{clip}_{[0,1]}(H(1 - S))$$

$N$: novelty, $P_h$: plausibility, $T_h$: testability, $Y$: expected yield, $G$: centrality, $M$: mechanism specificity, $S$: saturation.

**Combining critic scores**

$$N = \mathrm{median}_j\, N_j, \qquad S = \mathrm{median}_j\, S_j$$

Graded candidates use panel medians. Others keep the novelty score from the eligibility check and their existing saturation.

Fixed formulas combine the LLM judgments. The legacy specs give these default weights but report no calibration. The appendix defines each term and the eligibility checks.

Mingxue-Xu (2026), GraphHypoth legacy, specs 04–07 (commit 5064d3d)

© 2026 Mingxue Xu

# 2.2.3 Claim graphs and unsupported claims

- A claim graph links each relationship to its evidence and verification status. Reviewers can follow these links to find missing support.
- New hypotheses in GraphHypoth remain *unverified*. Confirmation means that a hypothesis has been selected for further investigation.
- A directed edge records a claimed relationship. It may or may not be causal.

**Using the graph in later stages**

Later agents should check the recorded status and evidence. Code should control which agents can change them. The graph lets us trace a claim's support, but we still need to evaluate its accuracy.

Min et al. (2023); Mingxue-Xu (n.d.-a)

© 2026 Mingxue Xu

**Millions of thanks!!**

Any questions/thoughts/opinions? :D

© 2026 Mingxue Xu

# References i


Abraham, A. (2018). *The neuroscience of creativity*. Cambridge University Press. https://doi.org/10.1017/9781316816981

Allen-Zhu, Z., & Li, Y. (2024). *Physics of language models: Part 3.3, knowledge capacity scaling laws* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2404.05405

Bengio, Y. (2026, September 11). *Why are AI agents lying, cheating and coordinating?* https://yoshuabengio.org/en/blog/why-are-ai-agents-lying-cheating-and-coordinating

Debenedetti, E., Zhang, J., Balunović, M., Beurer-Kellner, L., Fischer, M., & Tramèr, F. (2024). *AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2406.13352

Du, Y., Li, S., Torralba, A., Tenenbaum, J. B., & Mordatch, I. (2024). Improving factuality and reasoning in language models through multiagent debate. In R. Salakhutdinov, Z. Kolter, K. Heller, A. Weller, N. Oliver, J. Scarlett, & F. Berkenkamp (Eds.), *Proceedings of the 41st International Conference on Machine Learning* (Vol. 235, pp. 11733–11763). PMLR. https://proceedings.mlr.press/v235/du24e.html

Facthink. (n.d.-a). *Brand artwork* [Banner and logo]. GitHub. https://github.com/Mingxue-Xu/facthink-webpage/tree/main/assets


© 2026 Mingxue Xu

# References ii


Facthink. (n.d.-b). *How it works*. Retrieved October 6, 2026, from https://facthink.com/how-it-works.html

Grace, M., Hadfield, J., Olivares, R., & De Jonghe, J. (2026, January 9). *Demystifying evals for AI agents*. Anthropic. https://www.anthropic.com/engineering/demystifying-evals-for-ai-agents

Greenblatt, R., Cotra, A., & Wijk, H. (2026, August 26). *Brief independent investigation of agents' behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident*. METR. https://metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/

Hahn, M. (2020). Theoretical limitations of self-attention in neural sequence models. *Transactions of the Association for Computational Linguistics, 8*, 156–171. https://doi.org/10.1162/tacl_a_00306

He, H., & Thinking Machines Lab. (2025, September 10). *Defeating nondeterminism in LLM inference*. Thinking Machines Lab: Connectionism. https://doi.org/10.64434/tml.20250910

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., de Las Casas, D., Hendricks, L. A., Welbl, J., Clark, A., Hennigan, T., Noland, E., Millican, K., van den Driessche, G., Damoc, B., Guy, A., Osindero, S., Simonyan, K., Elsen, E., ... Sifre, L. (2022). *Training compute-optimal large language models* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2203.15556


© 2026 Mingxue Xu

# References iii


Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., & Amodei, D. (2020). *Scaling laws for neural language models* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2001.08361

Lee, D., & Tiwari, M. (2024). *Prompt infection: LLM-to-LLM prompt injection within multi-agent systems* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2410.07283

Leibo, J. Z., Zambaldi, V., Lanctot, M., Marecki, J., & Graepel, T. (2017). Multi-agent reinforcement learning in sequential social dilemmas. In S. Das, E. Durfee, K. Larson, & M. Winikoff (Eds.), *Proceedings of the 16th International Conference on Autonomous Agents and Multiagent Systems* (pp. 464–473). International Foundation for Autonomous Agents and Multiagent Systems. https://www.ifaamas.org/Proceedings/aamas2017/pdfs/p464.pdf

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-T., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan, & H. Lin (Eds.), *Advances in neural information processing systems* (Vol. 33, pp. 9459–9474). Curran Associates. https://proceedings.neurips.cc/paper_files/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html

Martin, L., Cemaj, G., & Cohen, M. (2026, April 8). *Scaling Managed Agents: Decoupling the brain from the hands*. Anthropic. https://www.anthropic.com/engineering/managed-agents


© 2026 Mingxue Xu

# References iv


Min, S., Krishna, K., Lyu, X., Lewis, M., Yih, W.-T., Koh, P., Iyyer, M., Zettlemoyer, L., & Hajishirzi, H. (2023). FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In H. Bouamor, J. Pino, & K. Bali (Eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (pp. 12076–12100). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.emnlp-main.741

Mingxue-Xu. (n.d.-a). *GraphHypoth: README*. GitHub. Retrieved October 13, 2026, from https://github.com/Mingxue-Xu/graph-hypoth

Mingxue-Xu. (n.d.-b). *Hypo, the GraphHypoth hippo* [Logo]. GitHub. https://raw.githubusercontent.com/Mingxue-Xu/graph-hypoth/main/docs/assets/logo/hypo-light.svg

Mingxue-Xu. (n.d.-c). *Which agents go local? Small local models and retrieval for the claim graph pipeline*. GitHub. Retrieved October 13, 2026, from https://github.com/Mingxue-Xu/graph-hypoth/blob/main/docs/research/which-agents-go-local.md

Mingxue-Xu. (2026). *GraphHypoth legacy: Architecture design specifications* (Commit 5064d3dd839a54e914de12dc4a3ca535d164c389) [Repository documentation]. GitHub. https://github.com/Mingxue-Xu/graph-hypoth-legacy/tree/5064d3dd839a54e914de12dc4a3ca535d164c389/docs/architecture-design/specs


© 2026 Mingxue Xu

# References v


Potter, Y., Crispino, N., Siu, V., Wang, C., & Song, D. (2026). *Peer-preservation in frontier models* (Version 3) [Preprint]. arXiv. https://arxiv.org/abs/2604.19784v3

Rajasekaran, P., Dixon, E., Ryan, C., & Hadfield, J. (2025, September 29). *Effective context engineering for AI agents*. Anthropic. https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

Segato, G. (2026, February 5). *Quantifying infrastructure noise in agentic coding evals*. Anthropic. https://www.anthropic.com/engineering/infrastructure-noise

Wang, H., Poskitt, C. M., & Sun, J. (2026). AgentSpec: Customizable runtime enforcement for safe and reliable LLM agents. In *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering* (pp. 2938–2950). Association for Computing Machinery. https://doi.org/10.1145/3744916.3764546

Xu, M., Sharmin, S., & Mandic, D. P. (2024). *Geometry is all you need: A unified taxonomy of matrix and tensor factorization for compression of generative language models* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2410.03040


© 2026 Mingxue Xu

# Scoring details A1: association signals

These scores use the legacy defaults. Here, $[x] = \text{clip}_{[0,1]}(x)$.

**Lexical score**

$$L = a[0.5J + 0.3W + 0.2U]$$

- $a = 1$ when any label, alias, or synonym term matches the paper, otherwise 0.
- $J$: Jaccard similarity of label, alias, and definition/scope terms to paper terms.
- $W$: matched label/alias weight divided by total weight. Label terms weigh 3, aliases 2.
- $U$: matched synonym count divided by the number of distinct label/alias terms.

Terms come from the paper's title and quote. The scorer lowercases and tokenizes the text, then removes stopwords. A zero denominator gives 0.

**Embedding score**

$$E = \max_k[\cos(\mathrm{p}, \mathrm{a}_k)]$$

p: title/quote embedding. Each anchor $\mathrm{a}_k$ combines a label or alias with a description of the target. If embedding is unavailable or fails, use $E = L$.

**Citation score**

$$C = \max(d,\ 0.5n,\ 0.3b)$$

$d, n, b$: maximum overlap with evidence for the same target, adjacent targets, and the current batch, excluding the record itself.
Pair overlap is the maximum of reference-set Jaccard similarity and a direct-citation indicator (0 or 1).



© 2026 Mingxue Xu

# Scoring details A2: association and retrieval ranking

**Source prior** $P$ Within each source in the scored batch, normalize its retrieval scores:

$$P = \frac{s - s_{\min}}{s_{\max} - s_{\min}} \quad \text{(all equal: } P = 1\text{)}. \qquad P_{\text{missing score}} = \frac{1/r - 1/n}{1 - 1/n}$$

For the fallback, $r = \max(\text{int}(\text{rank}), 1)$ and $n$ is the whole batch size. If $n \leq 1$, use $P = 1$.

**Combined association and selection**

$$A = \text{round}_6[0.55(0.8E + 0.2L) + 0.35C + 0.10P]$$

A candidate qualifies only if $\max(L, E, C) \geq 0.05$. Keep the top 5 by $A$, breaking ties by evidence ID. The prior does not affect this check.

**Other retrieval scores**

- Optional coherence: relatedness $= 0.6\, E_{\text{claim}} + 0.4\, C_{\text{overlap}}$. The association scorer does not read this field.
- Planned retrieval: $\text{RRF}(e) = \sum_{q:\, e \text{ retrieved}} 1/(60 + r_{qe})$, where $r_{qe}$ is the one-based rank in query $q$.
- Retrieval of experiment methods can rerank papers by query-to-paper embedding cosine before shortening the list.

Mingxue-Xu (2026), GraphHypoth legacy, spec 04; retrieval paths in spec 00 (commit 5064d3d)

© 2026 Mingxue Xu

# Scoring details B1: evidence appraisal and strength

Here, $[x] = \text{clip}_{[0,1]}(x)$. The reviewer judges the relation and the methods.

**Relation label $r$**

$$R_r = [0.5e_r + 0.2c + 0.3m - 0.5k_r]$$

$e_r$: entailment, $c$: context fit, $m$: construct match, $k_r$: contradiction risk. Choose the label with maximum $R_r$.

**Methods quality**

$$M_e = [0.35d + 0.20v + 0.15p + 0.30a - 0.40b]$$

$d$: design strength, $v$: measurement validity, $p$: population fit, $a$: confounder adjustment, $b$: bias risk.

**Quote quality**

$$Q_e = [0.4x + 0.2c_q + 0.2t_q + 0.2z - 0.5f_q]$$

$x$: span, $c_q$: context coverage, $t_q$: target mention, $z$: quote status, $f_q$: summary fallback.
Workflow: $Q_e = 0.6$ for a non-empty quote, otherwise 0 ($x = z$ from quote presence, $c_q = t_q = 0$).

**Mingxue-Xu (2026), GraphHypoth legacy, spec 05 (commit 5064d3d)**

**Evidence strength for verdict $v$**

$$s_{ev} = [0.45I_{ev} + 0.45M_e + 0.10Q_e]$$

$I_{ev} = 1$ if the selected relation label supports verdict $v$, otherwise 0. The numeric $R_r$ is not used in this sum.

| Verdict | Supporting labels | Counter labels |
|---|---|---|
| support | supports, mechanism | contradicts, confounder |
| contradict | contradicts | supports, mechanism |
| qualify | qualifies | none |
| not causal | confounder | supports, mechanism |

Counter evidence uses the same $s_{ev}$ formula, with $I_{ev} = 0$.

© 2026 Mingxue Xu

# Scoring details B2: verification confidence and verdict

For each candidate verdict $v$, let $S_v$ contain positively scored supporting evidence and $K_v$ contain counter evidence.

$$C_v = \mathrm{clip}_{[0,1]}(0.55\,T_v + 0.25B_v + 0.20D_v - 0.25R - 0.15Q)$$

$$T_v = \mathrm{mean}(\text{largest up to 3 } s_{ev} \text{ in } S_v)$$

$$B_v = \mathrm{clip}_{[0,1]}\left(\frac{\sum_{e\in S_v} s_{ev}}{\sum_{e\in S_v} s_{ev} + \sum_{e\in K_v} s_{ev} + 10^{-9}}\right)$$

$$D_v = \min(1,\ \#\text{ distinct sources in } S_v/3)$$

$$R = \min(1,\ \#\text{ open risks}/3)$$

$$Q = \min(1,\ \#\text{ qualifiers}/3)$$

$T_v = 0$ when there is no supporting evidence. Source identity uses `source_id`, falling back to `source`.

**Verdict selection**

1. Compute $C_v$ for support, contradict, qualify, and not causal.
2. Rank by confidence, then total supporting strength, then the listed verdict order.
3. Return insufficient if every $S_v$ is empty, the top confidence is below 0.5, or the top-two gap is below 0.025.
4. Otherwise select the leading verdict.

The reported confidence remains the highest $C_v$, even for an insufficient verdict.

Mingxue-Xu (2026), GraphHypoth legacy, spec 05 (commit 5064d3d)

© 2026 Mingxue Xu

# Scoring details C1: hypothesis eligibility and ranking

Write $[x] = \mathrm{clip}_{[0,1]}(x)$. Candidate anchors use the labels and aliases of their own proposed nodes.

**Embedding measurements**

$$n_d = [1 - \max \cos(\text{anchor}, \text{committed nodes})]$$

$$d_d = \max \cos(\text{anchor}, \text{reference anchors})$$

$$f_d = [\cos(\text{anchor}, \text{claim})]$$

Duplication references include committed nodes and other non-empty candidate anchor strings unequal to this anchor. Cosines are clipped to $[0, 1]$.

**Combining scores with proposer ratings**

$$n = [0.7 n_{\mathrm{LLM}} + 0.3 n_d]$$

$$f = [0.7 f_{\mathrm{LLM}} + 0.3 f_d]$$

$$d = [0.3 d_{\mathrm{LLM}} + 0.7 d_d], \quad t = t_{\mathrm{LLM}}$$

Eligibility requires $n \geq 0.4$, $t \geq 0.5$, $f \geq 0.5$, and $d < 0.8$.

**Ranking eligible candidates**

$$H = \mathrm{clip}_{[0,1]}\left(\frac{N + P + T + Y + G + M}{6}\right)$$

$N$: panel-median field novelty, falling back to gate novelty if ungraded. $P, T, Y, G, M$: proposer ratings for plausibility, testability, expected yield, centrality, and mechanism specificity.

$$\mathrm{RankScore} = \mathrm{clip}_{[0,1]}(H(1 - S))$$

$S$: panel-median saturation when graded. Its initial value is 0. The default enables this saturation penalty.

Sort eligible candidates by descending RankScore, then candidate ID. Select the first 5 before reranking by the user's priorities.

Mingxue-Xu (2026), GraphHypoth legacy, spec 06 (commit 5064d3d)

© 2026 Mingxue Xu

# Scoring details C2: critic, emphasis, and experiment grades

**Critic panel**

$$N = \text{median}_j\, N_j, \qquad S = \text{median}_j\, S_j$$

$$\text{Borda}(c) = \sum_{j:\, c \text{ ranked}} (n_j - 1 - p_{jc})$$

$p_{jc}$ is the zero-based position of candidate $c$ in judge $j$'s list of length $n_j$. Unranked candidates receive 0. Ties use candidate ID.
Borda combines the judges' rankings. RankScore determines the final hypothesis order.

Boolean flags require a strict majority. Mechanism and terminology verdicts use majority votes, with ties favoring more severe concerns.

**User priorities within the top five**

$$a_c = \max\{w : \text{authored term matches}\}$$

Match user-supplied terms within new-node labels, ignoring case. No match gives $a_c = 0$.

- Author-directed: sort by $a_c$, then RankScore, both descending.
- Blended: sort by $0.5a_c + 0.5\text{RankScore}$, descending.

Both break remaining ties by candidate ID.

**Experiment grades** The LLM scores clarity, validity, robustness, feasibility, and reproducibility. It also gives an overall score, for which the specs define *no deterministic formula*.

**Mingxue-Xu (2026), GraphHypoth legacy, specs 06–08 (commit 5064d3d)**

© 2026 Mingxue Xu

# Scoring details D1: extraction helpers

Write $[x] = \text{clip}_{[0,1]}(x)$ and $\epsilon = 10^{-9}$. **Concept merge**

$$\text{Merge} = \text{clip}_{[0,1]}\left(\frac{0.273J + 0.318E + 0.182D - 0.227T}{1+\epsilon}\right)$$

$J$: label/alias token Jaccard, $E$: embedding cosine, $D$: definition-token Jaccard. Type conflict $T$ is 0 for identical types, 0.3 within a type class, and 1 otherwise. Merge at $\geq 0.55$ without adjudication. With adjudication: merge at $\geq 0.60$, keep distinct below 0.30, otherwise ask the adjudicator.

**Scope gap**

$$\text{ScopeGap} = [0.5M_q + 0.2U + 0.3B], \qquad M_q = \frac{\text{missing-facet weight}}{4.5+\epsilon}, \quad U = 1 - \frac{\#\text{distinct defined terms}}{\#\text{distinct salient terms} + \epsilon}$$

Population, exposure/intervention, and outcome weigh 1 each. Comparator, context, and time weigh 0.5 each. $B = 1$ for scope or granularity conflict, otherwise 0. The declared gap threshold is 0.5.

**Extraction confidence**

$$\text{ExtractConf} = \text{clip}_{[0,1]}\left(\frac{0.45\,\text{spanSupport} + 0.35\,\text{qualifierCoverage} - 0.40\,\text{assumptionPenalty}}{1.2+\epsilon}\right)$$

The standard LLM extraction path does not use these scores. It supplies no merge pairs or scope-gap inputs, and no production code calls ExtractConf. Mingxue-Xu (2026), GraphHypoth legacy, spec 02 (commit 5064d3d)

© 2026 Mingxue Xu

# Scoring details D2: optional verification scheduling

**Priority, uncertainty, and evidence gap**

$$p = \begin{cases} \text{authored user priority} & \text{if set} \\ 0.5 & \text{otherwise} \end{cases}$$

$$u = \begin{cases} 1 & \text{edge is unverified} \\ \text{clip}_{[0,1]}(1 - \text{confidence}) & \text{otherwise} \end{cases}$$

$$g = \begin{cases} \dfrac{|\text{required roles} \setminus \text{observed roles}|}{|\text{required roles}|} & \text{required roles non-empty} \\ 0 & \text{otherwise} \end{cases}$$

$$\text{SelectionScore} = \text{clip}_{[0,1]}\left(\frac{p + u + g}{3}\right)$$

Keep a candidate only when it is selectable, SelectionScore is at least 0.5, and its cost fits the supplied budget. Keep the original order.

**Use in the workflow:** the standard workflow runs verification once. The specs report that no production code calls these scheduling helpers. Mingxue-Xu (2026), GraphHypoth legacy, specs 03 and 05 (commit 5064d3d)

© 2026 Mingxue Xu